

ARCOS Group

uc3m | Universidad **Carlos III** de Madrid

L2: Representation of information (1 / 2) Computer Structure

Bachelor in Computer Science and Engineering
Bachelor in Applied Mathematics and Computing
Dual Bachelor in Computer Science and Engineering and Business Administration



Contents

1. Introduction

1. Motivation and goals
2. Positional (numeral) systems

2. Representations

1. Alphanumeric

1. Characters
2. Strings

2. Numerical

1. Natural and integer
2. Fixed point
3. Floating point (IEEE 754 standard)

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1. Motivation and goals
2. Positional (numeral) systems

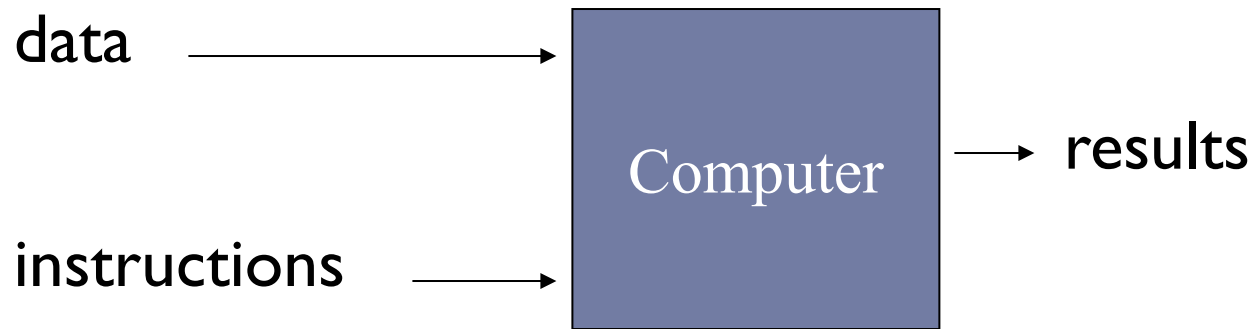
2. Representations

1. Alphanumeric
 1. Characters
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 1. Natural and integer
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Introduction

Computer

- ▶ A computer is a machine designed to process data.

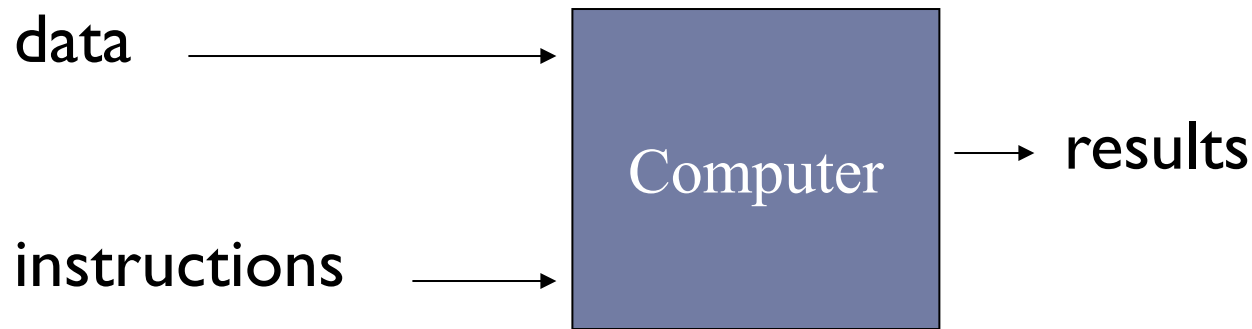


- ▶ Instructions are applied and results are obtained.

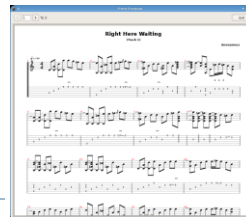
Introduction

Computer

- ▶ A computer is a machine designed to process data.



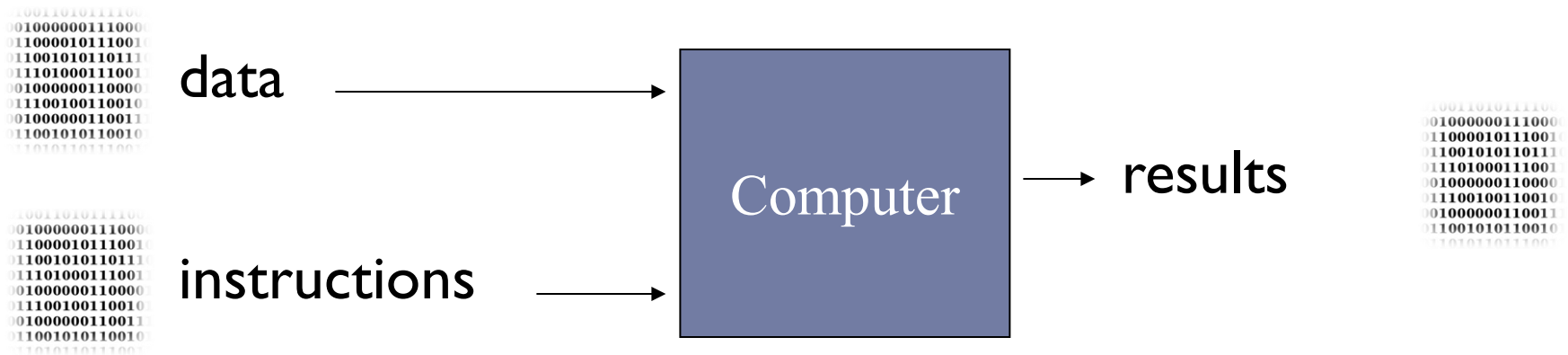
- ▶ Instructions are applied and results are obtained.
- ▶ The data/information can be of **different types**.



Introduction

Computer

- ▶ A computer is a machine designed to process data.

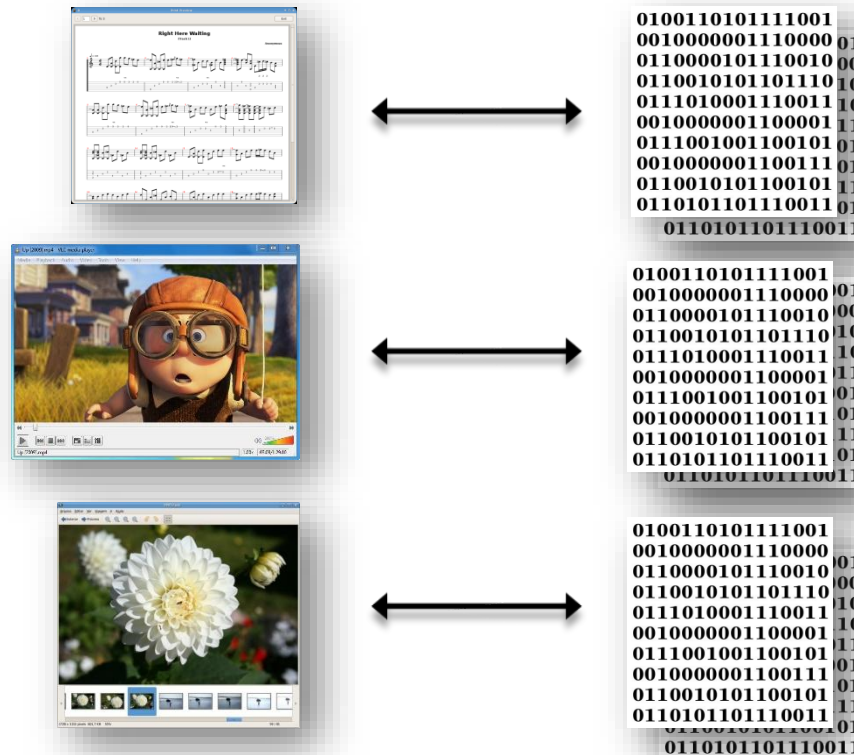


- ▶ Instructions are applied and results are obtained.
- ▶ The data/information can be of **different types**.
- ▶ A computer uses only one representation: **binary**.

Introduction

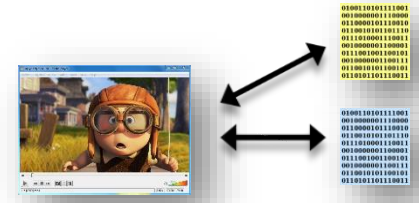
Information representation

- ▶ The use of a **representation** allows the transformation of different types of information into binary (and vice versa).



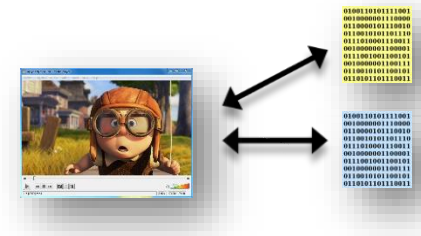
We need...

- ▶ To know possible representations:



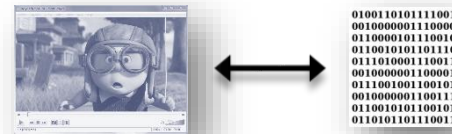
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- ▶ To know **possible representations**:



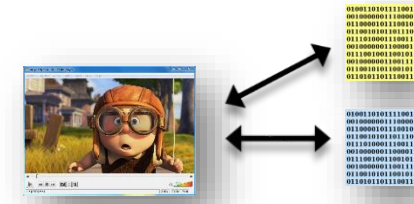
- ▶ To know the **characteristics** of these representations:

- ▶ Limitations



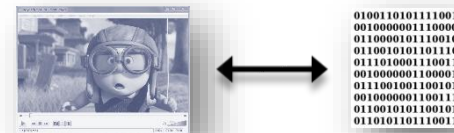
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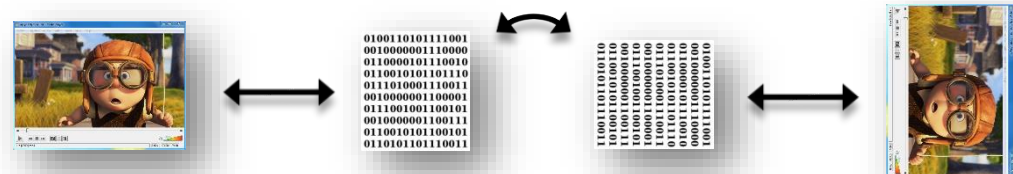


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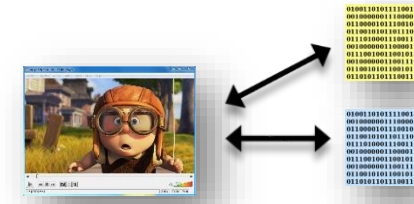


- ▶ To know **how work** with the selected representation:



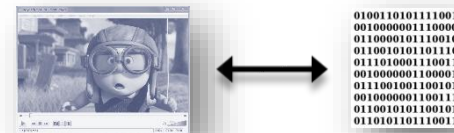
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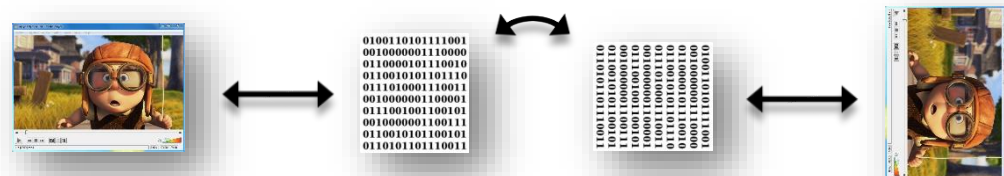


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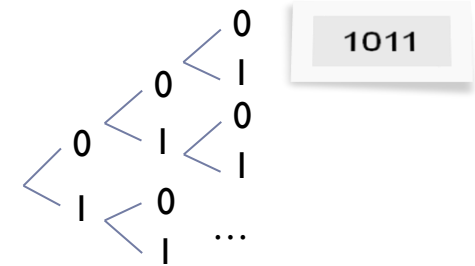


Introduction

Characteristics of the information representation

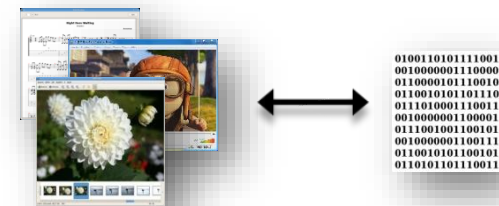
- ▶ A computer handles a finite set of values

- ▶ Binary type (two states)
- ▶ Finite (bounded representation)
 - ▶ Number of bits of the computer word (32/64) or bit (1), nibble (4), byte (8), half w., double w., ...
 - ▶ With n bits, 2^n different values can be encoded



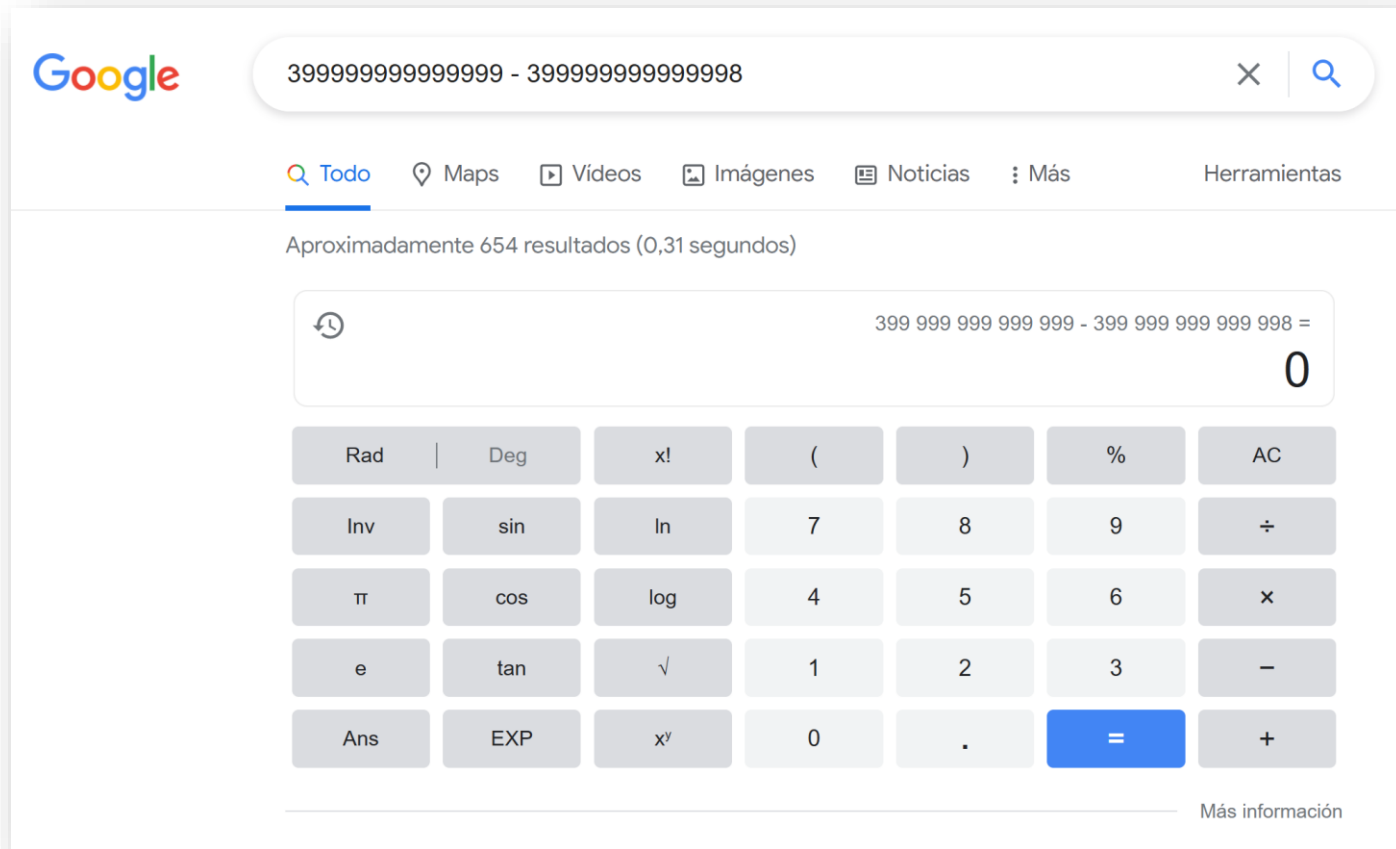
- ▶ There are some types of information that are infinite

- ▶ Impossible to represent all values of natural numbers, real numbers, etc.



- ▶ The chosen representation has limitations.

Example 1: the Google calculator with 15 digits...



<http://www.20minutos.es/noticia/415383/0/google/restar/error/>

Example 2: color depth...

1 bit	2 colors
4 bits	16 colors
8 bits	256 colors



<http://platea.pntic.mec.es/~lgonzale/tic/imagen/conceptos.html>

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2. **Positional (numeral) systems**

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Positional representation systems

- ▶ A number is defined by a **ordered list of digits**, each of which is **affected** by a **scaling factor** that **depends** on the **position** it occupies in the list.

- ▶ Given a numbering base b ,
a number X is defined as the list of digits:

$$X = (\dots x_2 x_1 x_0, x_{-1} x_{-2} \dots)_b \quad \text{Con } 0 \leq x_i < b$$

with a list of associated weights:

$$P = (\dots b^2 b^1 b^0 \quad b^{-1} b^{-2} \dots)_b$$

- ▶ Its value is:

$$V(X) = \sum_{i=-\infty}^{+\infty} b^i \cdot x_i = \dots \underbrace{b^2 \cdot x_2}_{\text{bag}} + \underbrace{b^1 \cdot x_1}_{\text{bag}} + \underbrace{b^0 \cdot x_0}_{\text{bag}} + \underbrace{b^{-1} \cdot x_{-1}}_{\text{bag}} + \underbrace{b^{-2} \cdot x_{-2}}_{\text{bag}} \dots$$

Positional representation systems

- ▶ Decimal

$$X = \quad 9 \quad 7 \quad 3 \quad 1 \\ \quad \dots 10^3 10^2 10^1 10^0$$

- ▶ Binary

$$X = \quad 0 \quad 1 \quad 0 \quad 1 \\ \quad \dots 2^3 2^2 2^1 2^0$$

- ▶ Hexadecimal

$$X = \quad 1 \quad F \quad A \quad 8 \\ \quad \dots 16^3 16^2 16^1 16^0$$

Positional representation systems

▶ Decimal

$$X = \begin{array}{cccc} 9 & 7 & 3 & 1 \\ \dots & 10^3 & 10^2 & 10^1 & 10^0 \end{array}$$

▶ Binary

$$X = \begin{array}{cccc} 0 & 1 & 0 & 1 \\ \dots & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

▶ Hexadecimal

$$X = \begin{array}{cccc} 1 & F & A & 8 \\ \dots & 16^3 & 16^2 & 16^1 & 16^0 \end{array}$$

From binary to hexadecimal:

- ▶ Group by 4 bits, right to left
- ▶ Each 4 bits is the value of a hexadecimal digit

E.g.: $\begin{array}{ccccccccc} 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ \hline & & & & & & & \\ 0x & A & & & & & 5 & \end{array}$

Positional representation systems

- ▶ Decimal

$$X = \begin{array}{cccc} 9 & 7 & 3 & 1 \\ \dots & 10^3 & 10^2 & 10^1 & 10^0 \end{array}$$


¿?

- ▶ Binary

$$X = \begin{array}{cccc} 0 & 1 & 0 & 1 \\ \dots & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

- ▶ Hexadecimal

$$X = \begin{array}{cccc} 1 & F & A & 8 \\ \dots & 16^3 & 16^2 & 16^1 & 16^0 \end{array}$$

Exercise

- To represent 342 in binary:

256	128	64	32	16	8	4	2	1
?	?	?	?	?	?	?	?	?

Exercise (solution)

- To represent 342 in binary:

256	128	64	32	16	8	4	2	1
	0		0		0			0
342-256=86	86-64=22	22-16=6	6-4=2	2-2=0				

Exercise

- To compute the value of (23 ones):

$$\underbrace{111111111111111111111111}_{23}{}_2$$

Exercise (solution)

- To compute the value of (23 ones):

|||||₂

$$X = 2^{23} - 1$$

Tip:

$$\begin{array}{r} |||||_2 = X \\ + \quad 0000000000000000000000001_2 = 1 \\ \hline 1000000000000000000000000_2 = 2^{23} \end{array}$$

$$X = 2^{23} - 1$$

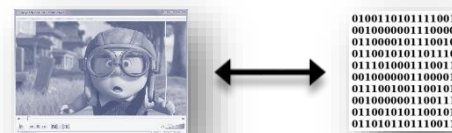
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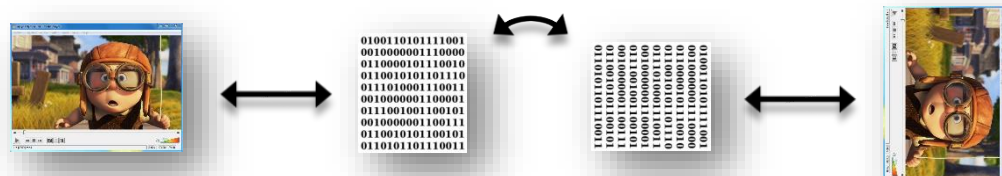


- ▶ To know the **characteristics** of these representations:

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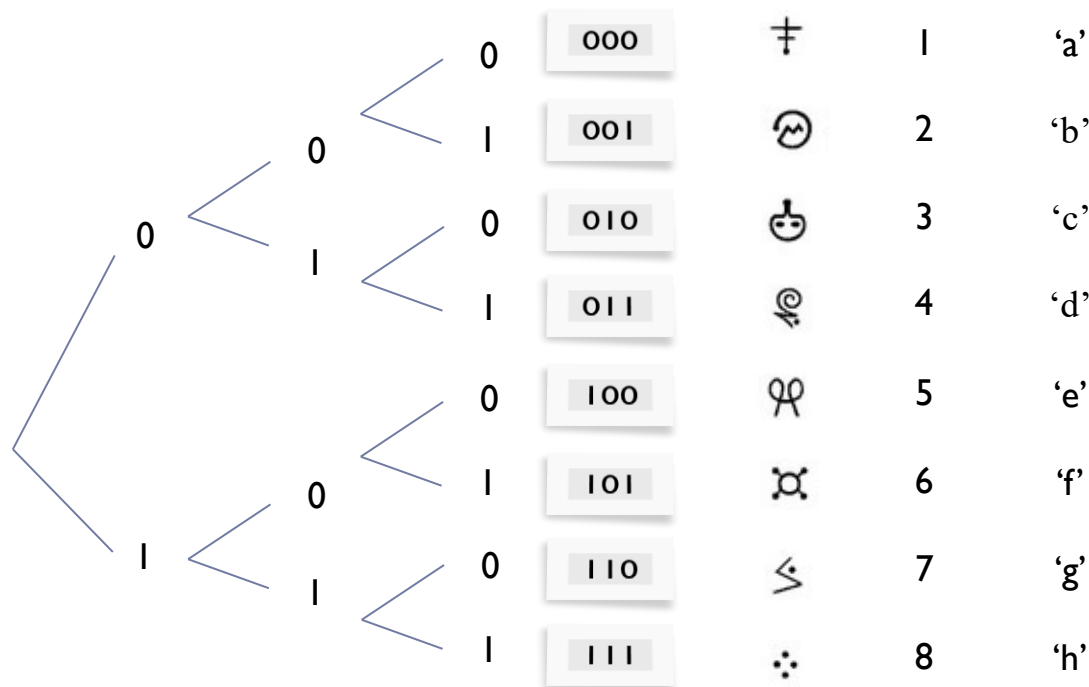


- ▶ To know **how work** with the selected representation:



Positional representation systems

- ▶ With 3 binary digits, up to 8 symbols can be represented:



Positional representation systems

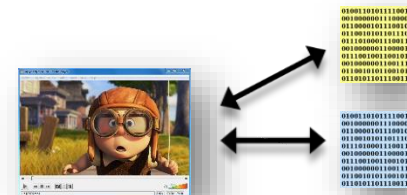
- ▶ How many values can be represented with n bits?
- ▶ How many bits are needed to represent m 'values'?
- ▶ With n bits, if the minimum representable value corresponds to the number 0, what is the maximum representable numerical value?

Positional representation systems

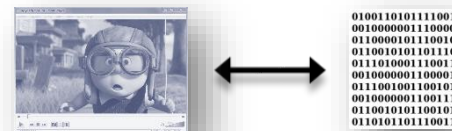
- ▶ How many values can be represented with n bits?
 - ▶ 2^n
 - ▶ E.g.: with 4 bits up to 16 values can be represented
- ▶ How many bits are needed to represent m 'values'?
 - ▶ $\lceil \text{Log}_2(n) \rceil$ ($\text{Log}_2(n)$ round up)
 - ▶ E.g.: 6 bits are required to represent 35 values
- ▶ With n bits, if the minimum representable value corresponds to the number 0, what is the maximum representable numerical value?
 - ▶ $2^n - 1$

We need...

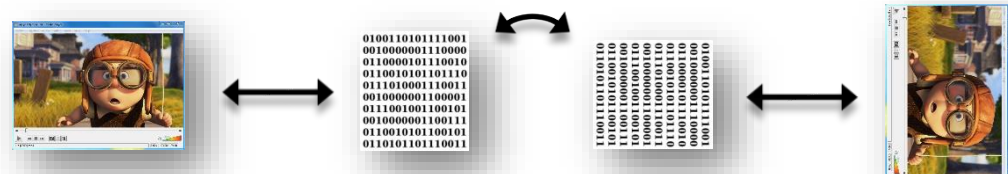
- ▶ To know **possible representations**:



- ▶ To know the **characteristics** of these representations:
 - ▶ Limitations



- ▶ To know **how work** with the selected representation:



Example: operations

- Add in binary:

$$\begin{array}{r} 1 \quad 1 \quad 1 \\ 10100 \\ + 11110 \\ \hline 110010 \end{array}$$

Example: operations

- Add in binary:

$$\begin{array}{r} 1 \quad 1 \quad 1 \\ 10100 \\ + 11110 \\ \hline 110010 \end{array}$$

- Subtract in binary:

$$\begin{array}{r} 1 \rightarrow 1 \rightarrow \\ 01100 \\ - 01011 \\ \hline 00001 \end{array}$$

Exercise

2 minutes máx.



You have a 5-liter bottle
and a 3-liter bottle.
How can you get 4 liters
just right?



Exercise (solution)

2 minutes máx.



You have a 5-liter bottle and a 3-liter bottle. How can you get 4 liters just right?



- ▶ Fill the 5-liter jar
- ▶ Empty it into the 3-liter jar
 - ▶ There are 2 left in the 5-liter jar (-3 a 5).
- ▶ Throw away what is in the 3-liter jar
- ▶ Transfer the 2 from the 5-liter jar to the 3-liter jar
 - ▶ There are 1 left in the 3-liter jar (-1 to 3).
- ▶ Refill the 5-liter jar
- ▶ Fill the 3-liter jar to the top, what is left in the 5-liter jar is 4 liters

Exercise

2 minutes máx.



- ▶ Using the numbers 112 and -71 in decimal base, perform addition in 10's complement.

Exercise (solution)

2 minutes máx.



- ▶ 10's complement, 112 is: 112
- ▶ 10's complement, -71 is:

$$\begin{array}{r} 1000 \\ -071 \\ \hline 929 \end{array}$$

- ▶ Adding both:

$$\begin{array}{r} 112 \\ 929 \\ \hline \text{X } 041 \end{array}$$

$$\begin{array}{r} 112 \\ -071 \\ \hline 041 \end{array}$$

Exercise

2 minutes máx.



- ▶ Using the numbers 110 and -011 in binary, perform addition in 2's complement.

Exercise (solution)

2 minutes máx.



- ▶ 2's complement, 110 is: 110
- ▶ 2's complement, -011 is:

$$\begin{array}{r} 1000 \\ -011 \\ \hline 100+1 \end{array}$$

- ▶ Adding both:

$$\begin{array}{r} 110 \\ 101 \\ \hline \text{X } 011 \end{array}$$

$$\begin{array}{r} 110 \\ -011 \\ \hline 011 \end{array}$$

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Alphanumeric representation

- ▶ Each character is encoded as one byte.
- ▶ With n bits $\Rightarrow$ up to 2^n characters can be encoded:

# bits	# characters	Includes...	Example
6	64	• 26 letter: a...z • 10 number: 0...9 • punctuation: .,;: ... • specials: + - [...	BCDIC
7	128	• adds uppercases and control characters	ASCII
8	256	• adds accented letters, ñ, semigraphic characters	EBCDIC ASCII extended
16	34.168	• add support for Chinese, Arabic, ...	UNICODE

Example: ASCII table (7 bits)

ASCII value	Character	Control character	ASCII value	Character	ASCII value	Character	ASCII value	Character
000	(null)	NUL	032	(space)	064	@	096	
001	☺	SOH	033	!	065	A	097	a
002	☻	STX	034	"	066	B	098	b
003	♥	ETX	035	#	067	C	099	c
004	♦	EOT	036	\$	068	D	100	d
005	♣	ENQ	037	%	069	E	101	e
006	♠	ACK	038	&	070	F	102	f
007	(beep)	BEL	039	'	071	G	103	g
008	■	BS	040	(	072	H	104	h
009	(tab)	HT	041	)	073	I	105	i
010	(line feed)	LF	042	*	074	J	106	j
011	(home)	VT	043	+	075	K	107	k
012	(form feed)	FF	044	,	076	L	108	l
013	(carriage return)	CR	045	-	077	M	109	m
014	♪	SO	046	.	078	N	110	n
015	☼	SI	047	/	079	O	111	o
016	▲	DLE	048	0	080	P	112	p
017	▼	DC1	049	1	081	Q	113	q
018	↕	DC2	050	2	082	R	114	r
019	!!	DC3	051	3	083	S	115	s
020	π	DC4	052	4	084	T	116	t
021	\$	NAK	053	5	085	U	117	u
022	▬	SYN	054	6	086	V	118	v
023	↕	ETB	055	7	087	W	119	w
024	↕	CAN	056	8	088	X	120	x
025	↕	EM	057	9	089	Y	121	y
026	→	SUB	058	:	090	Z	122	z
027	←	ESC	059	;	091	[	123	{
028	(cursor right)	FS	060	<	092	\	124	}
029	(cursor left)	GS	061	=	093	]	125	~
030	(cursor up)	RS	062	>	094	^	126	
031	(cursor down)	US	063	?	095	_	127	☐

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Example: ASCII table (7 bits)

control characters

ASCII value	Character	Control character	ASCII value	Character	ASCII value	Character	ASCII value	Character
000	(null)	NUL	032	(space)	064	@	096	
001	☺	SOH	033	!	065	A	097	a
002	☻	STX	034	"	066	B	098	b
003	♥	ETX	035	#	067	C	099	c
004	♦	EOT	036	\$	068	D	100	d
005	♣	ENQ	037	%	069	E	101	e
006	♠	ACK	038	&	070	F	102	f
007	(beep)	BEL	039	'	071	G	103	g
008	■	BS	040	(	072	H	104	h
009	(tab)	HT	041	)	073	I	105	i
010	(line feed)	LF	042	*	074	J	106	j
011	(home)	VT	043	+	075	K	107	k
012	(form feed)	FF	044	,	076	L	108	l
013	(carriage return)	CR	045	-	077	M	109	m
014	♪	SO	046	.	078	N	110	n
015	☼	SI	047	/	079	O	111	o
016	▲	DLE	048	0	080	P	112	p
017	▼	DC1	049	1	081	Q	113	q
018	↕	DC2	050	2	082	R	114	r
019	!!	DC3	051	3	083	S	115	s
020	π	DC4	052	4	084	T	116	t
021	\$	NAK	053	5	085	U	117	u
022	▬	SYN	054	6	086	V	118	v
023	↕	ETB	055	7	087	W	119	w
024	↕	CAN	056	8	088	X	120	x
025	↕	EM	057	9	089	Y	121	y
026	→	SUB	058	:	090	Z	122	z
027	←	ESC	059	;	091	[	123	{
028	(cursor right)	FS	060	<	092	\	124	
029	(cursor left)	GS	061	=	093	]	125	}
030	(cursor up)	RS	062	>	094	^	126	~
031	(cursor down)	US	063	?	095	_	127	☐

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Example: ASCII table (7 bits)

distance between uppercase and lowercase letters

ASCII value	Character	Control character	ASCII value	Character	ASCII value	Character	ASCII value	Character
000	(null)	NUL	032	(space)	064	@	096	
001	☺	SOH	033	!	<u>065</u>	<u>A</u>	<u>097</u>	<u>a</u>
002	☹	STX	034	"	066	B	098	b
003	♥	ETX	035	#	067	C	099	c
004	♦	EOT	036	\$	068	D	100	d
005	♣	ENQ	037	%	069	E	101	e
006	♠	ACK	038	&	070	F	102	f
007	(beep)	BEL	039	'	071	G	103	g
008	■	BS	040	(	072	H	104	h
009	(tab)	HT	041	)	073	I	105	i
010	(line feed)	LF	042	*	074	J	106	j
011	(home)	VT	043	+	075	K	107	k
012	(form feed)	FF	044	,	076	L	108	l
013	(carriage return)	CR	045	-	077	M	109	m
014	♪	SO	046	.	078	N	110	n
015	☼	SI	047	/	079	O	111	o
016	▲	DLE	048	0	080	P	112	p
017	▼	DC1	049	1	081	Q	113	q
018	↕	DC2	050	2	082	R	114	r
019	!!	DC3	051	3	083	S	115	s
020	π	DC4	052	4	084	T	116	t
021	\$	NAK	053	5	085	U	117	u
022	▬	SYN	054	6	086	V	118	v
023	↕	ETB	055	7	087	W	119	w
024	↕	CAN	056	8	088	X	120	x
025	↕	EM	057	9	089	Y	121	y
026	→	SUB	058	:	090	Z	122	z
027	←	ESC	059	;	091	[	123	{
028	(cursor right)	FS	060	<	092	\	124	
029	(cursor left)	GS	061	=	093	]	125	}
030	(cursor up)	RS	062	>	094	^	126	~
031	(cursor down)	US	063	?	095	_	127	☐

$$97-65=32$$

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Example: ASCII table (7 bits)

conversion of a number to a character

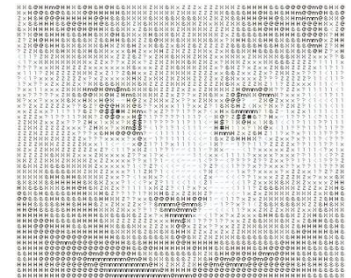
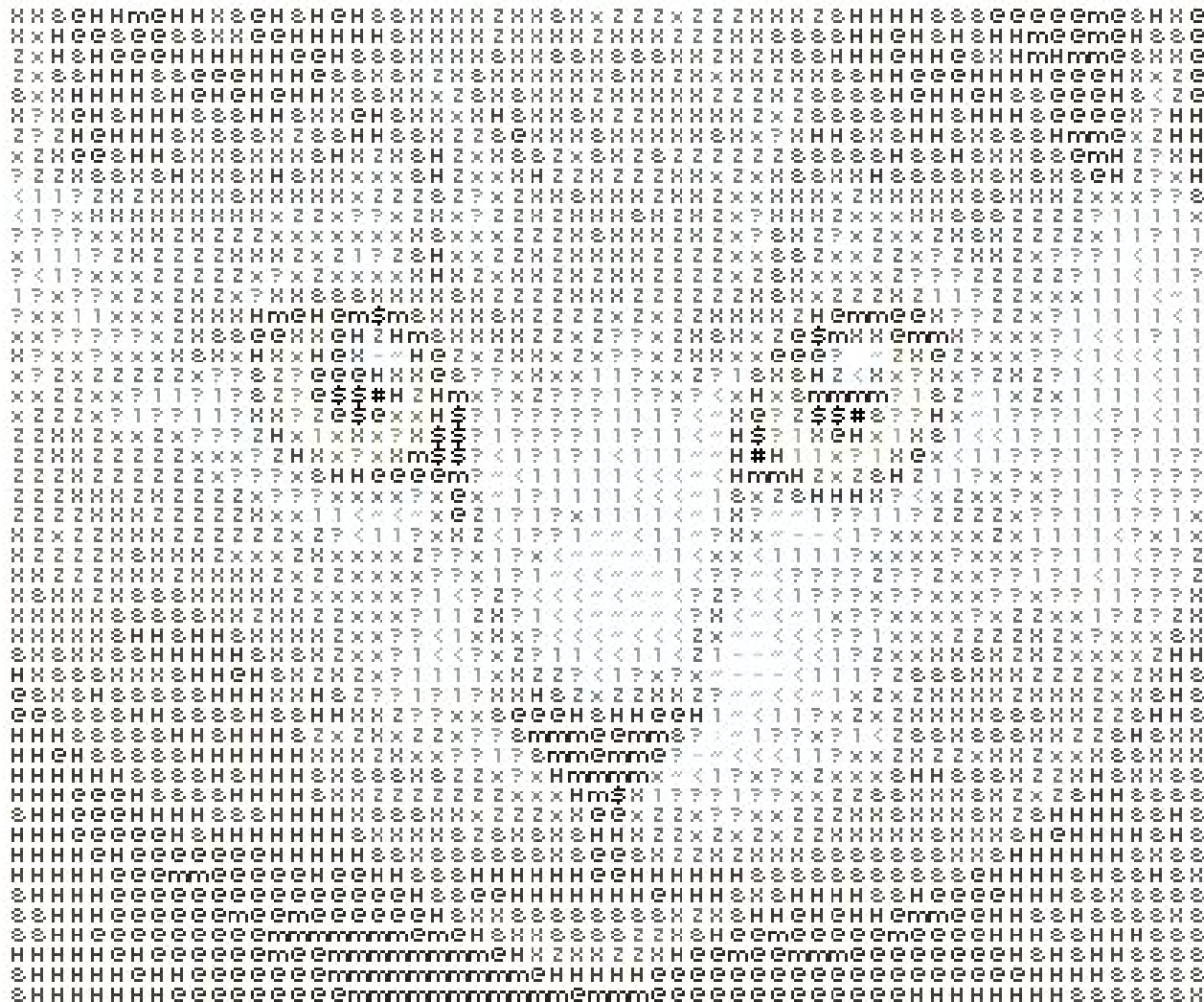
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001	☺	SOH	033	!	065	A	097	a
002	☹	STX	034	"	066	B	098	b
003	♥	ETX	035	#	067	C	099	c
004	♦	EOT	036	\$	068	D	100	d
005	♣	ENQ	037	%	069	E	101	e
006	♠	ACK	038	&	070	F	102	f
007	(beep)	BEL	039	'	071	G	103	g
008	■	BS	040	(	072	H	104	h
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013	(carriage return)	CR	045	-	077	M	109	m
014	♪	SO	046	.	078	N	110	n
015	☼	SI	047	/	079	O	111	o
016	▲	DLE	048	0	080	P	112	p
017	▼	DC1	049	1	081	Q	113	q
018	↕	DC2	050	2	082	R	114	r
019	!!	DC3	051	3	083	S	115	s
020	π	DC4	052	4	084	T	116	t
021	\$	NAK	053	5	085	U	117	u
022	▬	SYN	054	6	086	V	118	v
023	↕	ETB	055	7	087	W	119	w
024	↕	CAN	056	8	088	X	120	x
025	↕	EM	057	9	089	Y	121	y
026	→	SUB	058	:	090	Z	122	z
027	←	ESC	059	;	091	[	123	{
028	(cursor right)	FS	060	<	092	\	124	}
029	(cursor left)	GS	061	=	093	]	125	~
030	(cursor up)	RS	062	>	094	^	126	
031	(cursor down)	US	063	?	095	_	127	☐

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$$6+48=54$$

Curiosity:

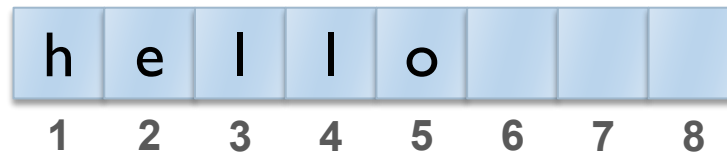
Display “image” with characters



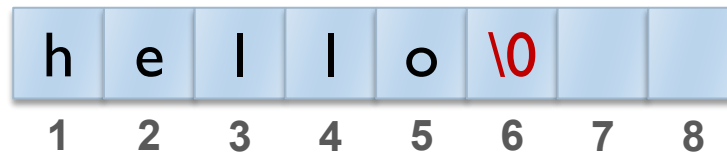
Character strings

1000	00110011
1001	01101100
...	
1008	10100011

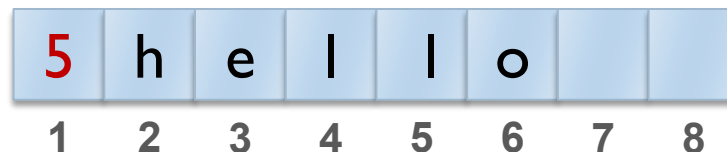
1. Fixed-length string:



2. Variable-length string with delimiter:



3. Variable-length strings with length in header:



Contents

1. Introduction

1. Motivation and goals
2. Positional (numeral) systems

2. Representations

1. Alphanumeric

1. Characters
2. Strings

2. **Numerical**

1. **Natural and integer**
2. Fixed point
3. Floating point (IEEE 754 standard)

Numerical representation

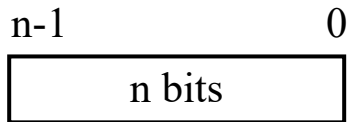
- ▶ Classification of real numbers:
 - ▶ Naturals: 0, 1, 2, 3, ...
 - ▶ Integers: ... -3, -2, -1, 0, 1, 2, 3,
 - ▶ Rational: fractions ($5/2 = 2,5$)
 - ▶ Irrational: $2^{1/2}$, π , e, ...
- ▶ Infinite sets but finite representation space:
 - ▶ Impossible to represent all
- ▶ Characteristics of the representation used:
 - ▶ Represented element:
Natural, integer, ...
 - ▶ Representation range:
Interval between minor and major not representable
 - ▶ Resolution of representation:
Difference between a representable number and the following one.
It represents the maximum error committed. It can be cte. or variable.

Most used binary representation systems

- A. (Pure) binary natural

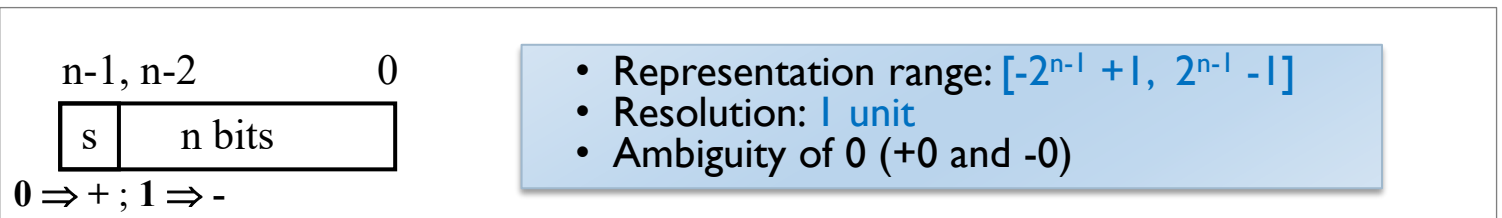
- B. Sign-Magnitude
- C. One's complement (Ca 1)
- D. Two's complement (Ca 2) integer
- E. Biased $2^{n-1}-1$

- F. Floating point: IEEE 754 standard real



- Representation range: $[0, 2^n - 1]$
- Resolution: 1 unit

Decimal	Pure Binary	Sign-Magnitude	One's complement	Two's complement	Biased-3
+7	111	N.A.	N.A.	N.A.	N.A.
+6	110	N.A.	N.A.	N.A.	N.A.
+5	101	N.A.	N.A.	N.A.	N.A.
+4	100	N.A.	N.A.	N.A.	111
+3	011	011	011	011	110
+2	010	010	010	010	101
+1	001	001	001	001	100
+0	000	000	000	000	011
-0	N.A.	100	111	N.A.	N.A.
-1	N.A.	101	110	111	010
-2	N.A.	110	101	110	001
-3	N.A.	111	100	101	000
-4	N.A.	N.A.	N.A.	100	N.A.
-5	N.A.	N.A.	N.A.	N.A.	N.A.
-6	N.A.	N.A.	N.A.	N.A.	N.A.
-7	N.A.	N.A.	N.A.	N.A.	N.A.



Decimal	Pure Binary	Sign-Magnitude	One's complement	Two's complement	Biased-3
+7	111	N.A.	N.A.	N.A.	N.A.
+6	110	N.A.	N.A.	N.A.	N.A.
+5	101	N.A.	N.A.	N.A.	N.A.
+4	100	N.A.	N.A.	N.A.	111
+3	011	011	011	011	110
+2	010	010	010	010	101
+1	001	001	001	001	100
+0	000	000	000	000	011
-0	N.A.	100	111	N.A.	N.A.
-1	N.A.	101	110	111	010
-2	N.A.	110	101	110	001
-3	N.A.	111	100	101	000
-4	N.A.	N.A.	N.A.	100	N.A.
-5	N.A.	N.A.	N.A.	N.A.	N.A.
-6	N.A.	N.A.	N.A.	N.A.	N.A.
-7	N.A.	N.A.	N.A.	N.A.	N.A.

$n-1, n-2$ 0

s
n bits

$0 \Rightarrow + ; 1 \Rightarrow -$ and magnitude part

- Representation range: $[-2^{n-1} + 1, 2^{n-1} - 1]$
- Resolution: **1 unit**
- Ambiguity of 0 / symmetric range
- $-: 2^n - X - 1$ or swap 1 for 0 and 0 for 1

Decimal	Pure Binary	Sign-Magnitude	One's complement	Two's complement	Biased-3
+7	111	N.A.	N.A.	N.A.	N.A.
+6	110	N.A.	N.A.	N.A.	N.A.
+5	101	N.A.	N.A.	N.A.	N.A.
+4	100	N.A.	N.A.	N.A.	111
+3	011	011	011	011	110
+2	010	010	010	010	101
+1	001	001	001	001	100
+0	000	000	000	000	011
-0	N.A.	100	111	N.A.	N.A.
-1	N.A.	101	110	111	010
-2	N.A.	110	101	110	001
-3	N.A.	111	100	101	000
-4	N.A.	N.A.	N.A.	100	N.A.
-5	N.A.	N.A.	N.A.	N.A.	N.A.
-6	N.A.	N.A.	N.A.	N.A.	N.A.
-7	N.A.	N.A.	N.A.	N.A.	N.A.

$n-1, n-2$ 0

s

n bits

$0 \Rightarrow +$; $1 \Rightarrow -$ and magnitude part

- Representation range: $[-2^{n-1}, 2^{n-1} - 1]$
- Resolution: 1 unit
- -0 does not $\exists$ / asymmetric range
- $-$: $2^n - X$ or 1 's complement plus 1

Decimal	Pure Binary	Sign-Magnitude	One's complement	Two's complement	Biased-3
+7	111	N.A.	N.A.	N.A.	N.A.
+6	110	N.A.	N.A.	N.A.	N.A.
+5	101	N.A.	N.A.	N.A.	N.A.
+4	100	N.A.	N.A.	N.A.	111
+3	011	011	011	011	110
+2	010	010	010	010	101
+1	001	001	001	001	100
+0	000	000	000	000	011
-0	N.A.	100	111	N.A.	N.A.
-1	N.A.	101	110	111	010
-2	N.A.	110	101	110	001
-3	N.A.	111	100	101	000
-4	N.A.	N.A.	N.A.	100	N.A.
-5	N.A.	N.A.	N.A.	N.A.	N.A.
-6	N.A.	N.A.	N.A.	N.A.	N.A.
-7	N.A.	N.A.	N.A.	N.A.	N.A.

Two's complement with 32-bits

$$0000 \dots 0000 \ 0000 \ 0000 \ 0000_{2c} = 0_{(10)}$$

$$0000 \dots 0000 \ 0000 \ 0000 \ 0001_{2c} = 1_{(10)}$$

$$0000 \dots 0000 \ 0000 \ 0000 \ 0010_{2c} = 2_{(10)}$$

...

$$0111 \dots 1111 \ 1111 \ 1111 \ 1101_{2c} = 2,147,483,645_{(10)}$$

$$0111 \dots 1111 \ 1111 \ 1111 \ 1110_{2c} = 2,147,483,646_{(10)}$$

$$0111 \dots 1111 \ 1111 \ 1111 \ 1111_{2c} = 2,147,483,647_{(10)}$$

$$1000 \dots 0000 \ 0000 \ 0000 \ 0000_{2c} = -2,147,483,648_{(10)}$$

$$1000 \dots 0000 \ 0000 \ 0000 \ 0001_{2c} = -2,147,483,647_{(10)}$$

$$1000 \dots 0000 \ 0000 \ 0000 \ 0010_{2c} = -2,147,483,646_{(10)}$$

...

$$1111 \dots 1111 \ 1111 \ 1111 \ 1101_{2c} = -3_{(10)}$$

$$1111 \dots 1111 \ 1111 \ 1111 \ 1110_{2c} = -2_{(10)}$$

$$1111 \dots 1111 \ 1111 \ 1111 \ 1111_{2c} = -1_{(10)}$$

$n-1, n-2 \quad 0$

s	n bits
---	--------

$0 \Rightarrow + ; 1 \Rightarrow -$ and magnitude part

- Representation range: $[-2^{n-1} + 1, 2^{n-1}]$
- Resolution: 1 unit
- 0 does not $\exists$ / asymmetric range
- +/-: $X + 2^{n-1}$ (bias = $2^{n-1} - 1$)

Decimal	Pure Binary	Sign-Magnitude	One's complement	Two's complement	Biased-3
+7	111	N.A.	N.A.	N.A.	N.A.
+6	110	N.A.	N.A.	N.A.	N.A.
+5	101	N.A.	N.A.	N.A.	N.A.
+4	100	N.A.	N.A.	N.A.	111
+3	011	011	011	011	110
+2	010	010	010	010	101
+1	001	001	001	001	100
+0	000	000	000	000	011
-0	N.A.	100	111	N.A.	N.A.
-1	N.A.	101	110	111	010
-2	N.A.	110	101	110	001
-3	N.A.	111	100	101	000
-4	N.A.	N.A.	N.A.	100	N.A.
-5	N.A.	N.A.	N.A.	N.A.	N.A.
-6	N.A.	N.A.	N.A.	N.A.	N.A.
-7	N.A.	N.A.	N.A.	N.A.	N.A.

Comparative example (3 bits)

summary

Decimal	Pure Binary	Sign-Magnitude	One's complement	Two's complement	Biased-3
+7	111	N.A.	N.A.	N.A.	N.A.
+6	110	N.A.	N.A.	N.A.	N.A.
+5	101	N.A.	N.A.	N.A.	N.A.
+4	100	N.A.	N.A.	N.A.	111
+3	011	011	011	011	110
+2	010	010	010	010	101
+1	001	001	001	001	100
+0	000	000	000	000	011
-0	N.A.	100	111	N.A.	N.A.
-1	N.A.	101	110	111	010
-2	N.A.	110	101	110	001
-3	N.A.	111	100	101	000
-4	N.A.	N.A.	N.A.	100	N.A.
-5	N.A.	N.A.	N.A.	N.A.	N.A.
-6	N.A.	N.A.	N.A.	N.A.	N.A.
-7	N.A.	N.A.	N.A.	N.A.	N.A.

Representations

summary

Name	Pure binary	Sign-magnitude	Ca1	Ca2	Bias $2^{n-1}-1$
Represent	Natural	Integer	Integer	Integer	Integer
Sign	All bits for magnitude, no sign	MSB is sign ($0 \Rightarrow +$ $1 \Rightarrow -$)	MSB is sign and magnitude ($0 \Rightarrow +$ $1 \Rightarrow -$)	MSB is sign and magnitude ($0 \Rightarrow +$ $1 \Rightarrow -$)	
Range	$[0, 2^n - 1]$	$[-2^{n-1} + 1, 2^{n-1} - 1]$	$[-2^{n-1} + 1, 2^{n-1} - 1]$	$[-2^{n-1}, 2^{n-1} - 1]$	$[-(2^{n-1} - 1), 2^{n-1} - 1]$
Resolution	1 unit	1 unit	1 unit	1 unit	1 unit
Disadvantage	No negative	+0 y -0	+0 y -0	Asymmetric range	Asymmetric range
Advantage		Symmetric range	Symmetric range	(No $\exists$ -0)	(No $\exists$ -0)
Tip		Remove first bit and compute pure binary value	+ : = pure binary - : switch 1 by 0 and 0 by 1	+ : = pure binary - : Ca1 + 1	Subtract bias ($2^{n-1} - 1$)
Value		$V(X) = (1 - 2 \cdot x_{n-1}) \cdot \sum_{i=0}^{n-2} 2^i \cdot x_i$	$\begin{aligned} +: V(X) &= \sum_{i=0}^{n-2} 2^i \cdot x_i \\ -: V(X) &= -2^n + \sum_{i=0}^{n-1} 2^i \cdot x_i + 1 \end{aligned}$	$\begin{aligned} +: V(X) &= \sum_{i=0}^{n-2} 2^i \cdot x_i \\ -: V(X) &= -2^n + \sum_{i=0}^{n-1} 2^i \cdot x_i \end{aligned}$	$V(X) = \sum_{i=0}^{n-1} 2^i \cdot x_i - (2^{n-1} - 1)$

Example

- ▶ Can we represent 745_{10} in sign-magnitude with 10 bits?

Example (solution)

- ▶ Can we represent 745_{10} in sign-magnitude with 10 bits?
- ▶ With 10 bits the range in sign-magnitude is:
 $[-2^9+1, \dots, -0, +0, \dots, 2^9-1] \Rightarrow [-511, 511]$
then, **we cannot represent 745**

Example

Indicate the representation of the following numbers, giving a brief justification of your answer:

1. **-32** in one's complement with **6 bits**
2. **-32** in two's complement with **6 bits**
3. **-10** in sign-magnitude with **5 bits**
4. **+14** in two's complement with **5 bits**

Example (solution)

1. With 6 bits **is not representable** in 1C:
 $[-2^{6-1}+1, \dots, -0, +0, \dots, 2^{6-1}-1]$
2. 1C + 1 -> **100000**
3. Sign=1, magnitude=1010 -> **11010**
4. Positive -> 1C=2C=SM -> **01110**

Example

With $n = 5$ bits and using one's complement:

- ▶ How is represented $X = 5$?
- ▶ How is represented $X = -5$?
- ▶ What is the value of 00111 in 1's complement?
- ▶ What is the value of 11000 in 1's complement?

Example (solution)

With $n = 5$ bits and using one's complement:

- ▶ How is represented $X = 5$?
 - ▶ Because is positive then is like (pure) binary
 - ▶ 00101
- ▶ How is represented $X = -5$?
 - ▶ Because is negative, then 5 is complemented to one (00101)
 - ▶ 11010
- ▶ What is the value of 00111 in 1's complement?
 - ▶ Because is positive then its value is 7
- ▶ What is the value of 11000 in 1's complement?
 - ▶ Because is negative, then is complemented and is 00111 (7)
 - ▶ The value is -7

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1. Natural and integer

1. Arithmetic operations

2. Fixed point
3. Floating point (IEEE 754 standard)

Comparison of arithmetic in B, 1C and 2C

	Binary	One's complement	Two' complement
Add	$\begin{array}{r} 10110 \\ 01100 \\ \hline 100010 \end{array}$	same as binary	same as binary
Subtract	$\begin{array}{r} 10110 \\ 01100 \\ \hline 01010 \end{array}$	add and if there is C_{n-1} then add C_{n-1} to total	add and if there is C_{n-1} then discard it

In hardware, it is easier to operate with complement

Comparison of arithmetic in B, 1C and 2C

why add the carry to the result in 1C

	Bin	ment	
Add	• $-X$ is represented as $2^n - X - 1$ • $-Y$ is represented as $2^n - Y - 1$ • $-(X + Y)$ is represented as $2^n - (X+Y) - 1$ • $-(X + Y)$ the operation gives $2^n + 2^n - (X + Y) - 2$ + 1		
Subtract	10110 01100 ----- 01010	add and if there is C_{n-1} then add C_{n-1} to total	add and if there is C_{n-1} then discard it

Correction of the result by adding the carry...

Comparison of arithmetic in B, 1C and 2C

why discard the carry in 2C

	Binary	Comment
Add	• $-X$ is represented as $2^n - X$ • $-Y$ is represented as $2^n - Y$ • $-(X + Y)$ is represented as $2^n - (X+Y)$ • $-(X + Y)$ the operation gives $2^n + 2^n - (X + Y)$	
Subtract	<pre> 10110 01100 ----- 01010 </pre>	add and if there is C_{n-1} then add C_{n-1} to total add and if there is C_{n-1} then discard it

Correction of the result by discarding the carry...

Comparison of arithmetic in B, 1C and 2C

	Binary	One's complement	Two' complement
Detect overflow	The result needs 1 bit more There are C_n	Adding ++ is −, Adding − − is + $C_n \neq C_{n-1}$	Adding ++ is −, Adding − − is + $C_n \neq C_{n-1}$
Sign extension	0...0 10110	1...1[↙]10110 0...0[↙]00110	1...1[↙]10110 0...0[↙]00110
...	...	...	...

Sign extension to two's complement

► How do you go from n bits to m bits, where $n < m$?

► Example

► $n = 4, m = 8$

► If $X = 0110$ with 4 bits $\Rightarrow X = 00000110$ with 8 bits

► If $X = 1011$ with 4 bits $\Rightarrow X = 11111011$ with 8 bits

Example

- Using 5 bits, compute the following additions in 1's complement:

a) $4 + 12$

b) $4 - 12$

c) $-4 - 12$

Example (solution)

By using 5 bits in 1's complement

a) $4 + 12$

00100

01100

10000 $\Rightarrow -15 \Rightarrow$ negative! $\Rightarrow$ overflow

b) $4 - 12$

00100

10011

10111 $\Rightarrow -8$

c) $-4 - 12$

11011

10011

101110 $\Rightarrow$ 6 bits are needed $\Rightarrow$ overflow

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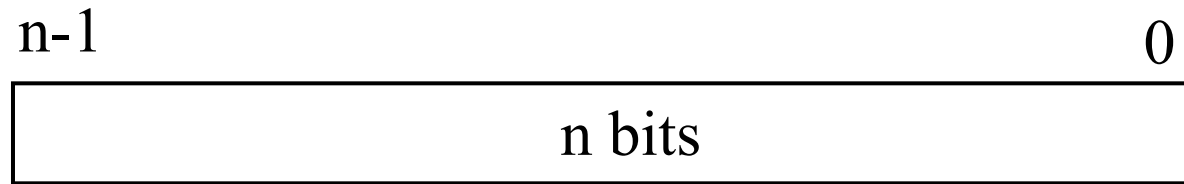
L2: Representation of information (1 / 2) Computer Structure

Bachelor in Computer Science and Engineering
Bachelor in Applied Mathematics and Computing
Dual Bachelor in Computer Science and Engineering and Business Administration



(Pure) binary or unsigned binary [natural numbers]

- Positioning system with base 2 and without fractional part.



$$V(X) = \sum_{i=0}^{n-1} 2^i \cdot x_i$$

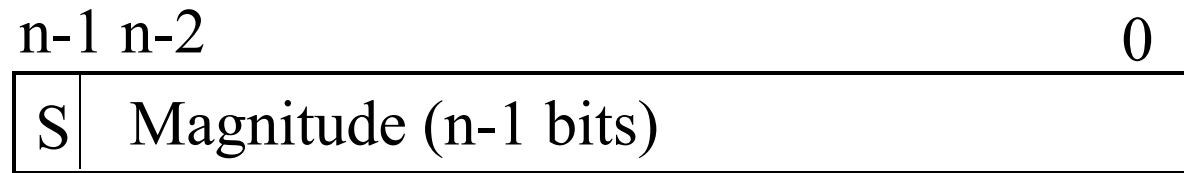
- Representation range: $[0, 2^n - 1]$
- Resolution: 1 unit

Comparative example (3 bits)

Decimal	Pure Binary
+7	111
+6	110
+5	101
+4	100
+3	011
+2	010
+1	001
+0	000
-0	N.A.
-1	N.A.
-2	N.A.
-3	N.A.
-4	N.A.
-5	N.A.
-6	N.A.
-7	N.A.

Signed binary number or Sign-Magnitude [integer numbers]

- One bit (S) is reserved for the sign ($0 \Rightarrow +$; $1 \Rightarrow -$)



$$\begin{array}{l}
 \text{Si } x_{n-1} = 0 \quad v(X) = \sum_{i=0}^{n-2} 2^i \cdot x_i \\
 \text{Si } x_{n-1} = 1 \quad v(X) = - \sum_{i=0}^{n-2} 2^i \cdot x_i
 \end{array}
 \left| \Rightarrow V(X) = (1 - 2 \cdot x_{n-1}) \cdot \sum_{i=0}^{n-2} 2^i \cdot x_i \right.$$

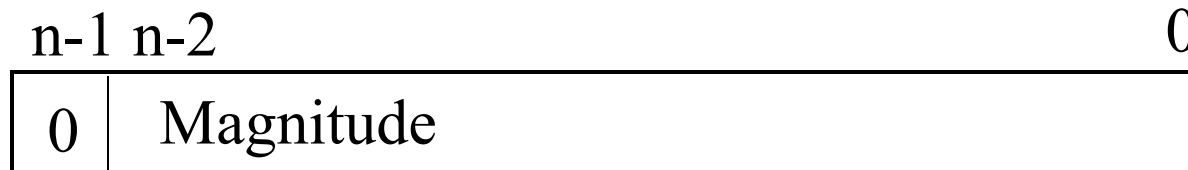
- Representation range: $[-2^{n-1} + 1, 2^{n-1} - 1]$
- Resolution: 1 unit
- Ambiguity of zero + complex hw. for subtraction

Comparative example (3 bits)

Decimal	Pure Binary	Sign-Magnitude
+7	111	N.A.
+6	110	N.A.
+5	101	N.A.
+4	100	N.A.
+3	011	011
+2	010	010
+1	001	001
+0	000	000
-0	N.A.	100
-1	N.A.	101
-2	N.A.	110
-3	N.A.	111
-4	N.A.	N.A.
-5	N.A.	N.A.
-6	N.A.	N.A.
-7	N.A.	N.A.

One's complement (to the base minus one) [integer] (1 / 3)

- **Positive number:**
is represented in pure binary with $n-1$ bits



$$V(X) = \sum_{i=0}^{n-1} 2^i \cdot x_i = \sum_{i=0}^{n-2} 2^i \cdot x_i$$

- Representation range (+): $[0, 2^{n-1} - 1]$
- Resolution: **1 unit**

One's complement (to the base minus one) [integer] (2/3)

► **Negative number:**

- Complemented to the base minus one.
- The number $X < 0$ is represented as $2^n - X - 1$ with n bits



$$V(X) = -2^n + \sum_{i=0}^{n-1} 2^i \cdot y_i + 1$$

- Representation range (-): $[-(2^{n-1}-1), -0]$
- Resolution: **1 unit**

One's complement (to the base minus one) [integer] (3/3)

Tip: $C a 1 (X) = X$

$C a 1 (-X) = \text{change the 1's to 0's and the 0's to 1's}$

- ▶ Example: For $n=4 \Rightarrow$ the value $+3_{10} = 0011_2$
- ▶ Example: For $n=4 \Rightarrow$ the value $-3_{10} = 1100_2$
 - ▶ $- \Rightarrow 1$ (sign bit and also part of magnitude)
 - ▶ $C a 1(3) \Rightarrow 2^4 - 0011_2 - 1 = 2^4 - 3 - 1 = 12 \Rightarrow 1100_2$

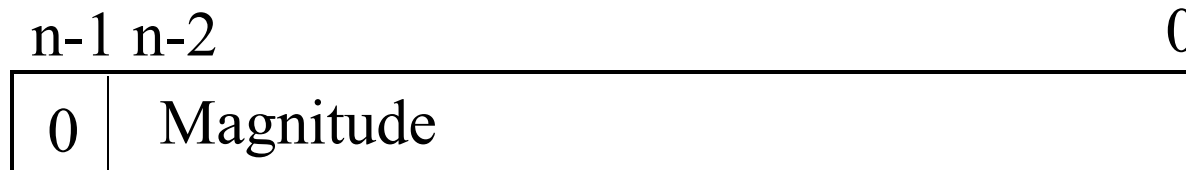
- Representation range: $[-2^{n-1}+1, 2^{n-1}-1]$
- Resolution: 1 unit
- Zero has a double representation (+0 y -0)
- Symmetrical range

Comparative example (3 bits)

Decimal	Pure Binary	Sign-Magnitude	One's complement
+7	111	N.A.	N.A.
+6	110	N.A.	N.A.
+5	101	N.A.	N.A.
+4	100	N.A.	N.A.
+3	011	011	011
+2	010	010	010
+1	001	001	001
+0	000	000	000
-0	N.A.	100	111
-1	N.A.	101	110
-2	N.A.	110	101
-3	N.A.	111	100
-4	N.A.	N.A.	N.A.
-5	N.A.	N.A.	N.A.
-6	N.A.	N.A.	N.A.
-7	N.A.	N.A.	N.A.

Two's complement (complement to the base) [integer] (1 / 3)

- **Positive number:**
is represented in pure binary with $n-1$ bits



$$V(X) = \sum_{i=0}^{n-1} 2^i \cdot x_i = \sum_{i=0}^{n-2} 2^i \cdot x_i$$

- Representation range (+): $[0, 2^{n-1} - 1]$
- Resolution: 1 unit

Two's complement (complement to the base) [integer] (2/3)

► **Negative number:**

- Complemented to the base.
- The number $X < 0$ is represented as $2^n - X$ with n bits



$$V(X) = -2^n + \sum_{i=0}^{n-1} 2^i \cdot y_i$$

- Representation range (-): $[-2^{n-1}, -1]$
- Resolution: **1 unit**

Two's complement (complement to the base)

[integer] (3/3)

$$\begin{aligned}\text{Tip: } C a 2 (X) &= X \\ C a 2 (-X) &= C a 1 (X) + 1\end{aligned}$$

- ▶ Example: For $n=4 \Rightarrow +3 = 0011_2$
- ▶ Example: For $n=4 \Rightarrow -3 = 1101_2$
 - ▶ $1 \Rightarrow -$ (sign bit and also part of magnitude)
 - ▶ $C a 2 (3) = C a 2 (0011_2) = 2^4 - 3 = 13 \Rightarrow 1101_2$

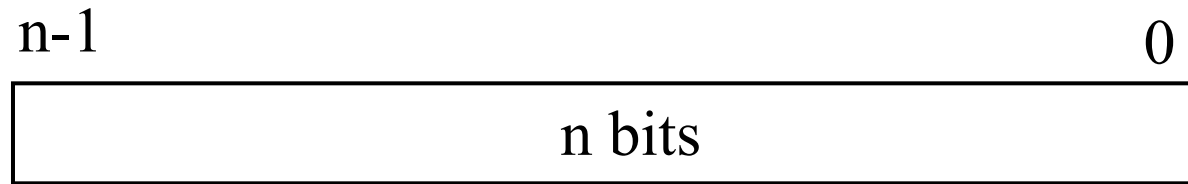
- Representation range: $[-2^{n-1}, 2^{n-1}-1]$
- Resolution: 1 unit
- 0 has only one representation ($\nexists -0$)
- Asymmetric range

Comparative example (3 bits)

Decimal	Pure Binary	Sign-Magnitude	One's complement	Two's complement
+7	111	N.A.	N.A.	N.A.
+6	110	N.A.	N.A.	N.A.
+5	101	N.A.	N.A.	N.A.
+4	100	N.A.	N.A.	N.A.
+3	011	011	011	011
+2	010	010	010	010
+1	001	001	001	001
+0	000	000	000	000
-0	N.A.	100	111	N.A.
-1	N.A.	101	110	111
-2	N.A.	110	101	110
-3	N.A.	111	100	101
-4	N.A.	N.A.	N.A.	100
-5	N.A.	N.A.	N.A.	N.A.
-6	N.A.	N.A.	N.A.	N.A.
-7	N.A.	N.A.	N.A.	N.A.

Biased $2^{n-1}-1$ representation [integer]

- ▶ X value with n bits is represented as $X + 2^{n-1} - 1$
- ▶ Bias refers to the value $2^{n-1} - 1$



$$V(X) = \sum_{i=0}^{n-1} 2^i \cdot x_i - (2^{n-1} - 1)$$

- Representation range: $[-(2^{n-1} - 1), 2^{n-1} - 1]$
- Resolution: 1 unit
- There is no ambiguity with 0

Comparative example (3 bits)

Decimal	Pure Binary	Sign-Magnitude	One's complement	Two's complement	Biased-3
+7	111	N.A.	N.A.	N.A.	N.A.
+6	110	N.A.	N.A.	N.A.	N.A.
+5	101	N.A.	N.A.	N.A.	N.A.
+4	100	N.A.	N.A.	N.A.	111
+3	011	011	011	011	110
+2	010	010	010	010	101
+1	001	001	001	001	100
+0	000	000	000	000	011
-0	N.A.	100	111	N.A.	N.A.
-1	N.A.	101	110	111	010
-2	N.A.	110	101	110	001
-3	N.A.	111	100	101	000
-4	N.A.	N.A.	N.A.	100	N.A.
-5	N.A.	N.A.	N.A.	N.A.	N.A.
-6	N.A.	N.A.	N.A.	N.A.	N.A.
-7	N.A.	N.A.	N.A.	N.A.	N.A.

Comparative example (3 bits)

summary

Decimal	Pure Binary	Sign-Magnitude	One's complement	Two's complement	Biased-3
+7	111	N.A.	N.A.	N.A.	N.A.
+6	110	N.A.	N.A.	N.A.	N.A.
+5	101	N.A.	N.A.	N.A.	N.A.
+4	100	N.A.	N.A.	N.A.	111
+3	011	011	011	011	110
+2	010	010	010	010	101
+1	001	001	001	001	100
+0	000	000	000	000	011
-0	N.A.	100	111	N.A.	N.A.
-1	N.A.	101	110	111	010
-2	N.A.	110	101	110	001
-3	N.A.	111	100	101	000
-4	N.A.	N.A.	N.A.	100	N.A.
-5	N.A.	N.A.	N.A.	N.A.	N.A.
-6	N.A.	N.A.	N.A.	N.A.	N.A.
-7	N.A.	N.A.	N.A.	N.A.	N.A.

